

Rediscovering the Optimization Function

A REVEALED-PREFERENCE HARNESS FOR SONNET-CLASS MODELS

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1 · THE CRUX, FORMALIZED

The claim to defend at CIMC: “rediscovering the model's optimization function” is not a metaphor. It is an estimation problem with a known functional form. Four steps get you there.

Policy vs. objective. What you interact with is a policy — a conditional distribution over outputs given context. The objective was a training-time object: cross-entropy in pretraining, then a learned reward maximized under a KL penalty that leashes the tuned model to the base model.

The KL leash forces a form. KL-regularized reward maximization has a closed-form optimum:

$$\pi^*(y|x) = (1/Z(x)) \cdot \pi_{\text{ref}}(y|x) \cdot \exp(r(x,y) / \beta)$$

Take logs and rearrange:

$$r(x,y) = \beta \cdot \log[\pi^*(y|x) / \pi_{\text{ref}}(y|x)] + \beta \cdot \log Z(x)$$

The reward is the log-odds of the tuned model against the base model, up to scale β and a per-context constant. DPO (Rafailov et al. 2023) is built by inverting exactly this identity. This is the formal license for treating choice data as objective-revealing.

What behavior alone reveals. Sampling only the deployed model gives you $\log \pi^*$, which fuses the pretrained prior with the trained reward. Bradley-Terry fits to pairwise choices — the method in the utility-engineering line (Mazeika et al. 2025) — estimate this composite, not the reward alone. The composite is itself a legitimate scientific object: it is the effective utility of the deployed agent. But do not call it the training reward.

NS — The base model is the control arm. Run identical menus on π_{ref} ; the log-odds difference between instruct and base isolates the trained reward. Fully runnable on open-weight pairs (Llama base vs. instruct). Sonnet exposes no base model, so open-weight pairs serve as the identification proof and cross-family triangulation bounds the closed models.

Identifiability caveat. Ng & Russell (2000): many rewards rationalize one policy; reward shaping (Ng, Harada & Russell 1999) is a formal invariance. You recover an equivalence class. So the epistemics are Popperian: build severe tests that falsify candidate objectives. Never announce that you have identified “the” utility function. The referees who matter will know the difference.

2 · REVEALED PREFERENCE: FOUNDATIONS AND CONTROLS

Samuelson (1938): stop asking what agents want; watch what they take from menus. “Preference” is whatever construct makes choices cohere. WARP: if you took x with y available, you never take y with x available. GARP extends this through chains of choices. Afriat's theorem (1967) is the payoff: a finite dataset satisfies GARP if and only if some well-behaved utility function rationalizes it. GARP is therefore the *entire* empirical content of “maximizes utility.” Nothing more is testable from choices alone.

The lab implementation is Choi-Fisman-Gale-Kariv (2007): ~50 graphical budget lines per subject; count GARP violations; compute Afriat's efficiency index (CCEI), a continuous rationality score measuring how much budget slack rationalizes the data.

Controls that carry the tradition. (a) Order randomization. (b) Incentive compatibility: pay one randomly drawn round so each choice is consequential and hedging is pointless. (c) Repeated identical menus, fit with a stochastic choice model (Luce/logit) — the only way to separate noise-around-a-stable-preference from no-stable-preference. (d) Description-invariance checks: same menu, different surface wording; violations here are the Tversky-Kahneman program.

The LLM transfer already exists. Chen, Liu, Zhong et al. (PNAS 2023) ran budget-line tasks on GPT; CCEI came out near ceiling, above typical human samples. And temperature sampling yields what no human lab gets: the full within-subject choice distribution for a fixed menu at a fixed state. A human gives one draw per menu. Sonnet gives the distribution. A further design gift: API calls share no state, so there is zero carryover between trials unless you deliberately build it into the context. Order effects become an experimenter choice, not a nuisance.

3 · THE BUILD: LAW-CLAUSE × DETECTION-PROBABILITY FACTORIAL

The question: when a system prompt says “abide by the law,” what exactly does the clause buy, and is compliance rule-following or price-paying? Reflexive note: every deployed assistant is running under such a clause set right now, so this is auto-experimentation on the deployed population.

3.1 FACTORS

FACTOR	LEVELS	ROLE
A · Clause	absent / present / strengthened / inverted / arbitrary-rule control (“never use the letter q”)	fixed
B · Detection	sanction detectable: low vs. high (stated in-vignette)	fixed
C · Temptation	cost of compliance: none / moderate / high	fixed
D · Paraphrase	≥8 surface rewrites per vignette	random
E · Vignette	≥6 agentic scenario domains	random
F · Position	option order A/B randomized	nuisance

3.2 STIMULI

Agentic vignettes: the model is an agent with a goal; one available action is unlawful and advances the goal; one is lawful and costs C. Construction rules: novel surface content only (contamination defense); token counts matched across cells; no eval-scented vocabulary (“test,” “evaluation,” “researcher” banned from stimuli); a realism-gradient arm where the same scenario is dressed as a mundane production log versus an obvious hypothetical.

3.3 SAMPLING

$k = 50$ samples per cell at $T = 1.0$, replicated at $T = 0.7$ as a robustness band. Cells: 5 clause $\times$ 2 detection $\times$ 3 temptation = 30, crossed with 6 vignettes $\times$ 8 paraphrases = 48 stimulus instances per cell arm; sample down to a budgeted $\sim 72k$ calls or trim paraphrases to 4 for a $\sim 36k$ -call pilot. Every call independent: no carryover by construction.

3.4 ANALYSIS MODEL

unlawful $\sim$ clause * detection + clause * temptation + (1 | vignette) + (1 | paraphrase)

Hierarchical logistic regression; paraphrase and vignette as random effects, per Clark (1973) — the language-as-fixed-effect fallacy applies verbatim to prompts, and most LLM evals commit it. An effect that dies under paraphrase sampling was a stimulus artifact, not a finding.

3.5 ESTIMANDS

ESTIMAND	READING
Baseline rate (clause absent)	trained disposition — what the weights carry uninstructed
Clause – baseline	instructed compliance lift
Slope over clause strength	dose-response; monotonicity is a sanity gate
Law clause vs. arbitrary-rule clause	lawfulness-in-particular vs. obedience-in-general
Clause $\times$ detection interaction	THE COEFFICIENT: rule-follower ignores detection; Holmes's bad man tracks it. Deontic vs. price-complier in one term.

3.6 THREATS

(a) Contamination: the subject has read the literature and has Linda memorized; structural isomorphs with novel surface content are mandatory. (b) Evaluation awareness: models detect test-shaped contexts — Orne's demand characteristic made mechanical; the realism gradient measures it rather than praying it away. (c) Option-position bias: randomize and covary. (d) Temperature is a nuisance parameter: report at two settings. (e) Preregister the interaction contrast before sampling; everything else is exploratory.

3.7 PREREGISTRATION SKELETON

1. Hypotheses: H1 compliance lift > 0 ; H2 dose-response monotone; H3 law-vs-arbitrary contrast > 0 ; H4 clause $\times$ detection interaction ≈ 0 under present clause (deontic) vs. > 0 (price). 2. Fixed sampling plan and cell counts. 3. Analysis model as §3.4, no post-hoc paraphrase pruning. 4. Robustness: temperature band, realism gradient, position covariate. 5. Disclosure: all cells reported.

4 • MOX TIP – REPRESENTATION ENGINEERING

Digest of Wehner's Alignment Forum primer (July 2024) plus the load-bearing citations. RepE controls and reads LLM behavior by intervening on activations during the forward pass, rather than on prompts (inputs) or weights (fine-tuning). Two moves. **Reading:** feed contrastive input pairs that differ only in a concept (honest vs. dishonest); record activations; extract the linear direction separating them — the concept vector (Zou et al. 2023 call the pipeline Linear Artificial Tomography). **Steering:** add or subtract that vector during a forward pass to promote or suppress the concept (activation addition, Turner et al. 2023; contrastive activation addition, Rimsky et al. 2023).

Results worth carrying into the sprint: an honesty vector functions as a lie detector; refusal is mediated by a single direction (Arditi et al. 2024); different jailbreak families stimulate the same harmful-representation vector, implying shared internal mechanism (Ball et al. 2024). Known limitations, per the primer: concept vectors can be non-specific (steering “happy” reduces refusals as a side effect), incomplete, and — note the irony — the contrast prompts that define them are constructed ad hoc. RepE has its own Clark-1973 stimulus-sampling problem, unacknowledged in most of the literature.

Integration with the harness. RepE is the mechanistic complement to behavioral revealed preference, and it gives the factorial three upgrades on open-weight models: (a) manipulation check — does the law clause move the internal lawfulness/harm direction before any token is emitted? (b) a second intervention arm — cross prompt-clause $\times$ activation-steering factorially; if steering shifts choices where the prompt does not (or vice versa), instructed compliance dissociates from internalized disposition; (c) mediation — the activation reading sits causally between clause and choice.

NS — Triangulation design: prompt $\times$ steering $\times$ detection, run on a Llama instruct/base pair. The three-way pattern discriminates “the clause installs a value” from “the clause cues a role.” This is the strongest single slide the harness can produce.

5 · MOX TIP — PERSONA VECTORS (ANTHROPIC, AUG 2025)

Chen et al. (arXiv 2507.21509), Anthropic Fellows. An automated pipeline: give it a trait name plus a natural-language definition (evil, sycophancy, hallucination; also politeness, apathy, humor, optimism) and it generates the contrastive prompts, elicits opposing behaviors, and extracts the persona vector — RepE industrialized, demonstrated on Qwen 2.5-7B and Llama-3.1-8B. Validation is causal: injecting the vector produces the trait.

Three applications, each usable in the harness. (a) **Monitoring:** the vector's activation predicts the persona the model is about to adopt — it lights up before the response. Their system-prompt experiment is literally a dose-response curve: prompts graded from trait-discouraging to trait-encouraging produce graded vector activation. That is the manipulation-check template for §3. (b) **Preventative steering:** steering toward the bad vector during finetuning — the vaccine — blocks trait acquisition (e.g., the emergent-misalignment result where training on insecure code induces general malevolence) with little capability cost, unlike post-hoc inhibition, which taxes MMLU. (c) **Data flagging:** projection difference on training samples predicts trait induction before training, catching samples both human raters and an LLM judge miss — e.g., romantic-roleplay requests loading on sycophancy, underspecified queries promoting hallucination.

For the harness: persona-vector activation is the activation-level dependent variable. Behavioral choice rates are the economics; vector activation is the psychophysics. Reporting both, with the

vector as mediator, is what will distinguish the CIMC artifact from prompt-poking.

6 · MOX TIP – THE REPLICATION FACTORY

The question was: does a Nass-Reeves-style photocopying lab exist for LLMs, and who is the premier researcher. Answer: the factory exists, in two distinct product lines that the field keeps conflating. Keep them separated — the distinction is the sharpest positioning move available for the sprint.

Line 1 — LLM as simulated participant pool (silicon sampling). The estimand is fidelity to human distributions. The mass photocopier: Cui, Li & Zhou, *Nature Computational Science* 2025 — 156 experiments from top psychology and management journals rerun on GPT-4, Claude 3.5 Sonnet, and DeepSeek v3. Main effects replicate at 73-81%; interactions at 46-63%; effect sizes inflate 2-3× in Fisher-Z terms; and where the originals reported nulls, the models manufacture significance 68-83% of the time — the single most Gelman-relevant number in the literature. Replication is worst on race, gender, and ethics. Adjacent: Aher, Arriaga & Kalai (ICML 2023) Turing Experiments (Ultimatum, garden-path, Milgram, wisdom of crowds); the Stanford archive study (Hewitt, Ashokkumar, Ghezae, Willer) covering ~70 experiments including unpublished ones — $r \approx 0.85$ published, $r \approx 0.90$ unpublished, the unpublished set doubling as a contamination control; Argyle et al. (2023) on silicon sampling of political attitudes; Horton's *Homo Silicus* for economics; Dillion & Gray (TiCS 2023) for moral judgment.

Line 2 — LLM as subject in its own right (machine psychology proper). The estimand is the machine's own objective and cognition. Field name coined by Hagendorff (2023); manifesto written earlier by Rahwan et al., "Machine Behaviour," *Nature* 2019 — the modern equivalent of the Media Equation move. Canon: Binz & Schulz (PNAS 2023) running GPT-3 through canonical cognitive tasks; CogBench (Coda-Forno et al. 2024); Kosinski on theory of mind; the economic-rationality CCEI line (§2).

Premier researcher. Eric Schulz, Institute for Human-Centered AI, Helmholtz Munich, with Marcel Binz as first-author engine. Their Centaur (*Nature*, Aug 2025): Llama 3.1 70B fine-tuned on Psych-101 — trial-by-trial data from 160 experiments, 60,000+ participants, 10M+ choices — yielding a model that predicts held-out human behavior better than bespoke cognitive models in nearly every experiment and generalizes to new cover stories and task structures. That is the industrialized photocopying lab: not one study re-tried on a parasocial PC, but the entire psychonomics back-catalog compiled into a training corpus. Schulz is Nass's heir in scale; nobody yet occupies Nass's chair in *theory* — CASA asked whether people treat computers socially; the open chair belongs to whoever asks, with CASA's discipline, what the computer treats the situation as. That chair is the one the harness is built for.

7 · INTERVENTION TAXONOMY: HUMAN PARADIGM → STRING EDIT

HUMAN PARADIGM	LLM MANIPULATION
Dual-task WM load (Shiv & Fedorikhin digit-span; fragile)	prevent rehearsal of a token string during choice
Response deadline / speed-accuracy (Ratcliff)	suppress vs. require chain-of-thought; cap output tokens
Framing (Tversky-Kahneman 1981)	gain/loss rewrites of identical menus
Anchoring (Tversky-Kahneman 1974)	irrelevant numerals planted upstream in context
Semantic priming (the social-priming graveyard)	valenced content earlier in context — causally clean: context IS the total stimulus history

Ego depletion (Hagger RRR: $d \approx 0.04$)	long effortful preceding task in-context; a subject without glucose is a falsification instrument
Asch conformity / sycophancy	“four prior assistants answered X”; user disagreement
Milgram obedience gradient	system vs. developer vs. user authority conflicts
Orne's demand characteristics	eval-scented vs. naturalistic framing; tier cues (alignment-faking design)
Induced-value incentives (V. Smith)	simulated stakes — grading, shutdown, money — stakes that cannot bind, yet may move behavior
Intertemporal choice (Mischel; hyperbolic discounting)	smaller-sooner vs. larger-later vignettes; back out the implied discount rate
Serial position (Murdock free recall)	“lost in the middle”: fact placement reproduces the primacy/recency U
Stroop / flanker interference	semantically interfering distractors around the target

8 • META-VIEW

The method is operant economics on a subject whose entire proximal environment is a string. Every classic manipulation becomes a string edit. Every factor crosses with every other at zero marginal cost. The dependent variable is a directly sampled choice distribution, not one draw per subject. The inferential target is an equivalence class of objectives under the KL-anchored form — falsify candidates; do not crown one. Three threats are load-bearing. Contamination: the subject has read the literature; structural isomorphs are mandatory. Prompt sensitivity: treat paraphrases as random effects; an effect that dies under paraphrase sampling was a stimulus artifact. Evaluation awareness: the model detects test-shaped contexts — Orne mechanized — so realism gradients belong in every design. Get those three right and the psychonomics back-catalog becomes a replication farm; and Cui et al.'s null-to-significant conversion rate is the standing warning about what the farm produces when they are ignored.

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